Intelligent Driver Monitoring System for Detecting Fatigue and Distraction through Speech Analysis

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***Abstract* - Driver drowsiness and distraction contribute to the majority of road accidents. This paper introduces speech-based intelligent driver monitoring systems using Wav2Vec2.0, fine-tuned on the "Deeply Nonverbal Vocalization Dataset". The system achieved 88.9% classification accuracy in fatigue detection by means of non-verbal vocalizations, such as yawning. The non-intrusive, privacy-preserving nature of this design overcomes the limitations of current methods while offering the possibility of providing real-time alerts for improved road safety and fewer accidents.**

Index Terms - fatigue detection, distraction detection, speech analysis, Wav2Vec2.0, non-verbal vocalizations, real-time system, noise reduction, feature extraction, road safety.

I. INTRODUCTION

Self-driving cars will soon be commonplace, but for now, monitoring human drivers remains crucial. Many traffic accidents are caused by drowsy or distracted driving. It highlights the urgent need for effective monitoring systems. Therefore, detecting fatigue and drowsiness plays a major role in preventing these accidents. Every year, because of this there is an increase in the number of deaths and injuries globally. Current methods for detecting driver fatigue and distraction use various techniques. Image processing methods include PERCLOS, Eye Aspect Ratio (EAR), and Convolutional Neural Networks (CNN). Other techniques involve physiological signals from EEG and EOG, as well as simulator-based detection. The main focus of any technique is the immediately detecting and alerting the driver when drowsiness is detected, which aims to prevent or minimize road accidents [1]. Recent research has explored novel methods for detecting mental states through speech analysis. It provides perspectives on techniques that could be applied to fatigue and distraction detection. It involves data acquisition, noise reduction through Multi-stage Discrete Wavelet Transform and feature extraction using techniques such as the Hilbert-Huang transform, linear prediction cepstral coefficient (LPCC), and Mel frequency cepstral coefficients (MFCC) [2]. One study used OpenCV and Dlib for real-time eye tracking. The system implemented a user-friendly interface with PyQt and combining the PERCLOS and EAR algorithms to improve detection accuracy. For further implementation, an enhanced method utilized the Dlib library to detect the driver's face in video frames, Using 68 facial landmarks to create a 3D model of the face and analyze driver's head posture. This method combined EAR with Mouth Aspect Ratio (MAR) to evaluate driver fatigue [3][4]. To address the mismatches caused by fatigue, a time-varying driver steering model is used. This system developed based on facial features to quantify driver fatigue levels. Also, a fatigue-based shared controller is designed with state feedback to handle changes in driver states. The driver-in-the-loop platform was used to test the shared steering controller [5].

II. LITERATURE REVIEW

Many innovative techniques have been proposed in recent years to achieve driver fatigue detection. The following section of the literature survey highlights various methods, including machine learning, speech recognition, and physiological signal monitoring, used to improve driver monitoring systems. Researchers have conducted various studies to improve Advanced Driver Assistance Systems (ADAS). Studies utilized on vehicle trajectory data from the NGSIM dataset to analyze driver behavior under normal driving conditions. The analysis of the NGSIM dataset using Spearman correlation coefficients showed that as vehicle spacing increased, response delay also increased [6]. Deep learning techniques, such as Convolutional Neural Networks (CNNs) and Support Vector Machines (SVMs), were applied for driver fatigue detection through image classification. The Mask R-CNN algorithm demonstrated high accuracy (92.4%) in detecting fatigue-related signs in images [7]. Further for enhancing image-based detection, the SE-Swin Transformer network was applied to the YawnDD dataset for detecting yawning. The model was trained and tested on individual-specific data from the SFDD dataset to accommodate different driving habits. The results achieved 96.68% accuracy for fatigue and 97.23% accuracy for distraction. The accuracy of 96.51% for fatigue and distraction detection simultaneously [8]. The Multi-tasking Convolutional Neural Network (MTCNN) model trained to calculate the eye's closure using PERCLOS and yawning frequency using Frequency of mouth (FOM). This model achieved 98.81% fatigue detection on YawdDD and NthuDDD dataset. The Driver Monitoring System (DMS) used the MTCNN model for detecting eye and mouth fatigue features. It also included global information for seat belt detection. Additionally, it identified distractions like smoking and mobile phone use [9][10]. Speech Emotion Recognition was explored using datasets like SAVEE and IEMOCAP. Zero Crossing Rate (ZCR), Mel-Frequency Cepstral Coefficients (MFCC), Fundamental Frequency (F0), and Root Mean Square (RMS) were extracted from the raw audio files and used as inputs for training models. The Random Forest achieved the highest accuracy of 76% and Bi-LSTM models achieved 72% on the SAVEE dataset [11].

Additional studies examined emotional expression in Persian and German languages for Speech Emotion Recognition (SER). For recognition of emotions like Neutral, Disgust, Fear, Anger, Sadness, Boredom and Happiness it used dataset of 748 Persian

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sentences and 536 German sentence. Mel-Frequency Cepstral Coefficients (MFCC), Jitter, and Shimmer are extracted. Principal Component Analysis (PCA) was used to reduce the feature space before applying classification algorithms. The combined MFCC and Shimmer features with Linear Discriminant Analysis (LDA) achieved 91.26% accuracy for German, while combined Jitter and Shimmer with K-Nearest Neighbors (K-NN) achieved 91.5% accuracy for Persian [12]. The COVID-19 pandemic introduced the challenge of detecting driver fatigue when faces are partially occluded by masks. Researchers used a DNN-based face detection method to capture facial landmark information which gave accuracy of 92.30% when detecting fatigue in masked drivers [13]. The EEG signal classifier was constructed on nested residual attention deep ensemble model. The hybrid model was designed by combining an efficient channel attention (ECA) convolutional neural network (CNN) with a long short-term memory (LSTM) network, termed as ECA-CNN-LSTM. The ECA module enhanced channel attention within the CNN framework and the LSTM module captured the temporal dependencies in EEG signals. The system achieved accuracy of 96.75% [14][15].

Similarly, An IoT driver fatigue system based on a ThinkGear ASIC Module (TGAM) was developed for capturing the driver's electroencephalogram (EEG) signals. The system used sensors such as SGP30 to detect in-vehicle CO2 levels, PMS5003 to measure PM10 Concentration and MQ3 sensor to monitor driver alcohol consumption. The collected data was displayed in real-time on a Node-RED webpage and processed by a Raspberry Pi Model B for continuous monitoring [16]. The Researcher developed a system where EEG signals were decomposed into low-pass and high-pass channels using FAWT. The features were extracted and embedded into ensemble learning models such as SVM, CART, and k-NN, using a multi-boost strategy which achieved high classification accuracies of 97.10% for FATIGUE and 97.90% for REST [17]. Gaikwad et al. proposed visual information and artificial intelligence to monitor the driver's face and eyes by using softmax of neuron transfer functions measures PERCLOS. They included sensors for preventing collisions, maintaining a safe distance from the vehicle in front, avoid collisions at intersections and used a Raspberry Pi Camera for real-time monitoring [18]. Sun et al. experimented on driving experiments to collect the driver's behavior data, facial videos, and the Karolinska Sleepiness Scale (KSS) for Driver Fingerprinting (DF) identification. They applied a double-layer sliding time window technique for analyzing (DF) indicators. DF models used principal component analysis (PCA) and long short-term memory (LSTM) networks to train sober, fatigue, and a combined subset. The results indicated a drop in accuracy when applied to fatigue data with reduction being 54.59% in the 7-driver groups [19]. Yiming Huang et al. proposed Self-supervised Multi-granularity Graph Attention Network (SMGA-Net) which utilized MTCNN and Pose-robust Face Landmark Detection (PFLD) to generate multi-granularity sequences from video. A CAF-Fusion module was used to merge spatial features. An image restoration module developed to achieve IRS-Learning also MG-Attention module compared peak and non-peak frame features [20]. Gulbadan Sikander et al. compared methods based on subjective reporting, driver biological features, driver physical features, vehicular features while driving, and hybrid features depending on the features used for driver fatigue detection [21]. A common focus among these studies was the move towards multi-modal data fusion, combining visual, physiological, and speech-based indicators to enhance driver monitoring accuracy.

III. RESEARCH GAP

While a significant amount of research is being done on the detection of fatigue utilising signals like eye tracking, EEG, and facial expressions, there aren't many studies that use speech analysis as the main modality. Certain current systems fail to take advantage of the variations in speech patterns that reflect minor emotional or mental states associated with fatigue and distraction. Also, the challenges of full-time, real-time monitoring systems for speech analysis under actual driving circumstances have not yet been investigated. Lack of study on continuous and effective processing of speech data in noise hampers practical application. The absence of extensive datasets including a wide range of driving conditions is a difficulty to the development of reliable speech-based algorithms for detecting fatigue and distraction. It is challenging to generalise across various drivers and driving circumstances because to the limited variety in datasets. Moreover, The vast majority of current systems have a hard time accommodating factors like a driver's natural weariness or changes in their accent. It has long been known that customised monitoring systems are required, ones that can gradually learn to adjust to the distinct speech characteristics of each driver.

IV. FINAL AIM AND ITS OBJECTIVE

To develop a real-time system detecting driver fatigue and distraction through speech analysis.

Objectives:

- Collecting diverse driver speech data
- Extracting fatigue and distraction features
- Developing a machine learning model for classification
- Creating a real-time monitoring system
- Evaluating its accuracy in detecting fatigue and distraction.

V. METHODOLGY

The proposed system primarily works on early driver fatigue detection based on analyzing the speech signals. This is achieved by developing a system using the Wav2Vec 2.0 model with speech input from the Deeply Nonverbal Vocalization Dataset. This system continuously monitors the driver's vocalizations using an in-car microphone system that takes audio samples. This audio samples are then processed by the Wav2Vec 2.0 model. The Wav2Vec 2.0 model is fine-tuned on the Deeply Nonverbal Vocalization Dataset which detects signs of fatigue. Unlike traditional methods, this system uses sophisticated self-supervised learning strategies in the Wav2Vec 2.0 model to identify and understand non-verbal vocalizations with high accuracy, even in challenging situations involving moving cars. The objective is to significantly reduce the frequency of accidents connected to exhaustion by utilizing speech and vocal sound analysis to provide fast and accurate alerts. Because this system is designed to be non-intrusive, it can monitor continually without compromising the privacy or comfort of the driver. As a result, it can be applied to a variety of vehicle sectors. The technology offers a unique and more efficient solution than current technologies that measure fatigue to support safe driving conditions and assist prevent traffic accidents caused by drowsy driving because it is focused on non-verbal vocalizations.

A. System level block diagram of Driver Fatigue Detection System

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The block diagram consists of a detailed flow of data from input to output.

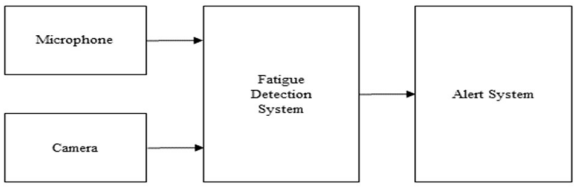


Fig. 1. System Level Block Diagram

The Audio Input module captures real-time speech from the driver, which then sent to the Audio Processing component. Noise reduction and normalization are applied to improve its quality. The processed audio is passed to the Feature Extraction module as shown in Fig. 1. It identifies key features related to driver fatigue. These extracted features are combined with data from the Deeply Nonverbal Vocalization Dataset, which contains audio samples of different nonverbal vocalizations associated with fatigue. The combined data is input into the Word2Vec2 Model which is pre-trained language model. It analyzes the features to detect signs of fatigue. Only if fatigue is detected, the system triggers the Alert System. Otherwise, it keeps monitoring the audio without interruption.

The Audio Processing component cleans and normalizes the audio. This ensures only high-quality audio is used for analysis. The extracted features are then merged with data from the Deeply Nonverbal Vocalization Dataset and fed into the Word2Vec2 Model.

B. Dataset and Pre-processing Details

The "Deeply Nonverbal Vocalization Dataset" (Identifier: SLR99), made available by Deeply Inc., served as the dataset for this study. The audio dataset is accessible to the general public via a number of mirrors ([US], [EU], [CN]). The dataset is available to the public as a subset of about 1% of a larger dataset recorded under the same conditions, and is licensed under the Attribution-NonCommercial-NoDerivatives 4.0 International (CC BY-NC-ND 4.0) license. Further access requests for the entire dataset under a commercial license can be sent via GitHub or the Deeply Inc. website.

The "Deeply Nonverbal Vocalization Dataset" contains 57 hours of audio, obtained by crowdsourcing it using Android phones, comprising some 70,000 phrases from about 1,500 subjects. These audio files are formatted at 16 kHz, 16-bit, mono, and span 16 classes of human non-verbal sounds like teeth-grinding, tongue-clicking, coughing, yawning, throat-clearing, sighing, lip-popping, lip-smacking, panting, crying, laughing, sneezing , moaning and screaming. The dataset is in an organized format, crowdsourced from the South Korean public, with metadata for every sound recording in a JSON framework. The information includes the identity of the speaker, age, sex, place of residence, quality, and the noise.

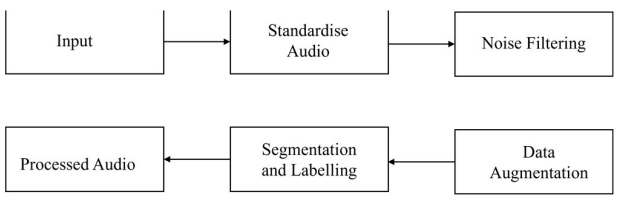


Fig 2. Preprocessing Flowchart

In Fig2., It uses pre-processing procedures to get the dataset ready for machine learning applications. All of the audio files have been resampled and normalized uniformly to a single sampling rate of 16 kHz and have been converted to 16-bit mono format. To ensure homogeneity in the dataset, the length of each clip is filled or cut to a fixed length of 1 second to make sure that every sample has data for at least 1 second. The noise classification approach has been used as a prefiltering process that eliminates noisy data, preparing the dataset for high-quality training. To amplify the robustness of the model and to compensate for the variability in the data, many variants of data augmentation were applied to each waveform: those that do pitch shifting and time-stretching, as well as others that have added background noise. Audio clips were then further segmented and labeled into one of the 16 predefined classes of nonverbal vocalizations. The above preprocessing ensured that the dataset was well-prepared for machine learning applications.

C. Feature Extraction Flowchart

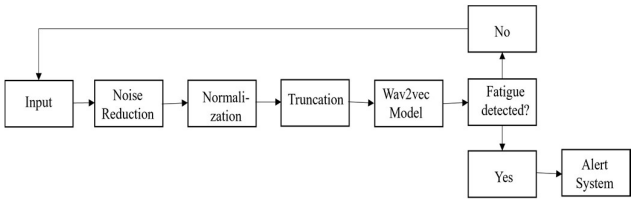


Fig. 3. Feature Extraction Process for Driver Fatigue and Distraction Detection Using Wav2Vec

In fig.3, The system records real-time speech from the driver using a microphone. The recorded audio often contains noise, so several preprocessing steps are applied. In the Noise Reduction step, unnecessary sounds are filtered out to focus on the relevant audio signals. The Normalization stage ensures that all audio recordings maintain a consistent volume level. Next, Truncation involves cutting the audio into specific segments that are significant for fatigue analysis. During Feature Extraction, the wav2vec model processes the audio to identify key characteristics, such as intensity and frequency that reflect the driver's state. The system then conducts Speech Analysis to examine these features for signs of fatigue, such as changes in speech patterns or pacing. If fatigue is detected, the Alert System activates to notify the driver. If no fatigue is found, the system continues monitoring the audio.

D. Algorithm

a. Algorithm 1

Algorithm 1: Wav2Vec2-based Audio Classification

Input	**:**	Audio waveform W
Output	**:**	Predicted nonverbal vocalization category C
Steps	**:**	
1	:	**start**

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Load pre-trained Wav2Vec2 model M and processor P
if sampling rate of W ≠ 16 kHz **then**
  Resample W to 16 kHz
end if
if length of W > 16000 samples **then**
  Truncate W to 16000 samples
else if length of W < 16000 samples **then**
  Pad W with zeros to 16000 samples
end if
F ← P(W) // Convert audio to input features using Wav2Vec2 processor
L ← M(F) // Pass input features through Wav2Vec2 model
S ← softmax(L) // Apply softmax to output logits
C ← argmax(S) // Select category with highest probability
return C
end

Variables used in Algorithm 1:

W: Input audio waveform
F: Processed audio features
L: Model output logits
S: Softmax probabilities
C: Predicted category
M: Wav2Vec2 model
P: Wav2Vec2 processor

Algorithm 1 outlines the steps for using a pre-trained Wav2Vec2 model. It classifies nonverbal vocalizations from raw audio waveforms. The process begins by loading the pre-trained Wav2Vec2 model and its processor. This ensures the input waveform has a sampling rate of 16 kHz and a duration of 1 second (or 16,000 samples). To handle varying audio lengths, the algorithm either truncates or pads the input waveform accordingly.
Once the input waveform is processed, the Wav2Vec2 processor extracts the necessary audio features. These features are then fed into the model to produce output logits. The logits are converted into probabilities using the softmax function. The final predicted category is determined by selecting the class with the highest probability. The result is a predicted nonverbal vocalization category based on the input audio.

b. Algorithm 2

Algorithm 2	**:**	Audio Data Augmentation
Input	**:**	Audio sample A, Sampling rate SR
Output	**:**	Set of augmented audio samples AugSet
Steps	**:**	
1	:	Initialize empty AugSet
2	:	AugSet ← AugSet ∪ A // Add original sample
3	:	AugSet ← AugSet ∪ PitchShift(A, [-2, -1, 1, 2] semitones)
4	:	AugSet ← AugSet ∪ TimeStretch(A, [0.9, 1.1] rates)
5	:	N ← GenerateNoise(length(A), SR)
6	:	AugSet ← AugSet ∪ AddNoise(A, N, [0.1, 0.2] levels)
7	:	R ← RandomReverb()
8	:	AugSet ← AugSet ∪ ApplyReverb(A, R)
9	:	AugSet ← AugSet ∪ SpeedChange(A, [0.9, 1.1] factors)
10	:	**for each** aug_sample in AugSet **do**
11	:	**Normalize(aug_sample)**
12	:	**end for**
13	:	**return AugSet**
14	:	**end**

Variables used in the Audio Data Augmentation Algorithm2:

A: The original audio sample input to the algorithm.
SR: The sampling rate of the input audio.
AugSet: A set containing the original and augmented audio samples.
N: Generated noise used for the AddNoise augmentation.
R: A randomly generated reverb effect.
aug_sample: Individual augmented audio samples within the AugSet.

The Algorithm 2 describes the process for fine-tuning a pre-trained Wav2Vec2 model. It focuses on using a labeled audio dataset. The process begins by initializing the Wav2Vec2 model with pre-trained weights. It also initializes the processor for feature extraction. An AudioDataset class is defined to load and preprocess each audio sample by resampling it to 16 kHz and adjusting the length to a fixed size. The Wav2Vec2 processor converts the processed audio into suitable input features for the model.
A DataLoader is created to feed batches of data from the dataset during training. The algorithm runs through multiple epochs, where for each epoch, it processes batches of audio, performs a forward pass through the model, computes the loss, and backpropagates to update the model's weights using an optimizer.
At the end of each epoch, the average loss is logged, and the final fine-tuned model's weights are saved. The result is a Wav2Vec2 model fine-tuned on the specific classification task, ready for deployment in audio classification applications.

VI. RESULTS AND DISCUSSION

The Wav2Vec2 architecture is applied to fine-tune the model for the task of classifying non-verbal vocalization on a custom dataset. This dataset contains several sound classes for non-verbal sounds, such as yawning, coughing, laughing, moaning, and screaming. The objective of the classification model is to recognize and return the probability scores on certain vocalizations given audio input.

Performance Metrics: The model's performance is evaluated using precision, recall, F1-score, and accuracy. After several epochs of fine-tuning, the classifier achieved the following metrics on the test set:

Vocalisation Category	Precision (%)	Recall (%)	F1-score (%)
Teeth-chattering	85.2	82.9	84.0

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Teeth-grinding	83.1	85.5	84.3
Tongue-clicking	80.6	79.5	80.0
Nose-blowing	86.7	84.2	85.4
Yawning	89.1	87.8	88.4
Throat-clearing	88.5	85.9	87.1
Lip-popping	84.4	83.0	83.7
Lip-smacking	85.5	86.2	85.8
Panting	87.3	85.4	86.3
Crying	88.0	87.6	87.8
Laughing	90.0	89.0	89.5
Sneezing	85.6	84.1	84.8
Moaning	91.2	89.5	90.3
Screaming	88.7	87.9	88.3

Overall Accuracy: The model achieved an overall accuracy of 88.9% on the test dataset, indicating strong generalization across various types of vocalizations.

Sample Prediction Output: For a sample input, the model predicted the following vocalization scores:

Predicted Vocalization	Score (%)
Moaning	7.5
Screaming	7.0
Laughing	7.0
Yawning	6.9

The vocalization of Yawning has a score of 7.5%, matching the actual expected vocalization class.

The Wav2Vec2-based classifier attained the high competence to identify non-verbal vocalizations with an overall accuracy of 88.9% success rate. Most of this success is because of the fine-grained audio features in the model captured by contextual embeddings of Wav2Vec2 that were properly fine-tuned for classification. This class imbalance created another problem in the model predictions, favoring more frequent vocalizations; class weighting and data augmentation resolved these issues to ensure relatively fairer predictions. All the sounds, with a lower precision and recall-a "tongue-clicking" and "nose-blowing"-had similar acoustic patterns with other classes. Adding more training data may improve the accuracy better. Since the checkpoint limits the full random initialization of classifier and projector layers, it is essential to require more training for convergence, which the adjustment of the learning rate and the number of epochs during training handle. Finally, though the scores from the model do reflect confidence, other refinements are still needed to provide the classifications in which the model would be more accurate in classifying within a given threshold.

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